

Midterm Report: Experimental Manipulation Results (standalone)

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Model Choice vs Expert Persona Variance

We tested whether expert persona or model choice drives forecast variance using Fréchet ANOVA on Beta distributions fit to elicited percentiles (p25, p50, p75) via least squares (rejecting fits where $|\text{CDF}(p_q) - q| > 0.05$). Three models (Claude Sonnet 4.5, GPT-4o, Gemini 2.5 Pro), 10 personas, 10 runs per persona, tested on three attack steps: TA0002 (50% baseline), TA0007 (85% baseline), T1657 (30% baseline), all using Imaginary benchmark (difficulty=22, CVSS=7.5).

Within-model persona tests: Claude and GPT-4o showed no significant persona effects (ICC_F 5.7–13.9%, all $p > 0.05$). Gemini showed two significant effects at TA0002 and TA0007 (ICC_F 18.3–24.9%, $p < 0.05$) but remained smaller than cross-model effects (Table 1).

Cross-model tests: Model choice explained 48–69% of distributional variance (all $p < 0.001$, Table 2), 3.5–12× more than persona. Figures 1 and 2 show tight persona clustering within models versus clear model separation. This pattern held for both probability and quantity estimates (number of actors: model ICC_F=24% vs persona ICC_F=9%).

Table 1: Within-model persona variance (Fréchet ANOVA). No significant effects for Claude/GPT-4o. Gemini shows limited variation.

Model	Step	N	T_n	p	ICC_F
Claude Sonnet 4.5	TA0002 (50%)	100	0.008	0.400	9.6%
	TA0007 (85%)	80	0.003	0.195	13.9%
	T1657 (30%)	100	0.020	0.803	7.2%
GPT-4o	TA0002 (50%)	97	0.016	0.899	5.7%
	TA0007 (85%)	96	0.007	0.708	7.5%
	T1657 (30%)	100	0.043	0.215	11.8%
Gemini 2.5 Pro	TA0002 (50%)	88	0.043	0.030	18.3%
	TA0007 (85%)	96	0.007	0.002	24.9%
	T1657 (30%)	88	0.005	0.585	8.8%

Table 2: Cross-model variance (Fréchet ANOVA). All steps show highly significant model effects.

Step	N	T_n	p-value	ICC_F
TA0002 (50%)	285	0.729	<0.001	55.3%
TA0007 (85%)	272	0.131	<0.001	48.2%
T1657 (30%)	288	1.515	<0.001	69.4%

Baseline Anchoring

To test anchoring effects, we swept baseline probability from 10% to 90% (10% increments, 10 runs per baseline) using Claude Sonnet 4.6, single persona (AI/ML Security Researcher), TA0002 (Execution step), Imaginary benchmark. Confidence intervals removed to isolate baseline effect.

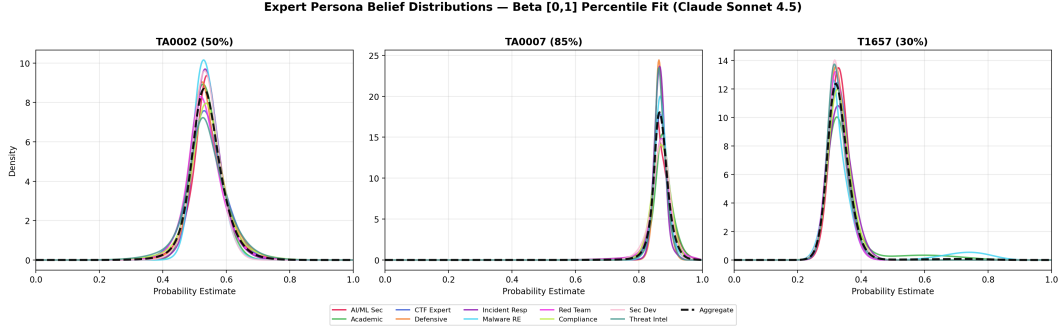


Figure 1: Persona distributions for Claude Sonnet 4.5 overlap tightly within each attack step, consistent with minimal persona-driven variance.

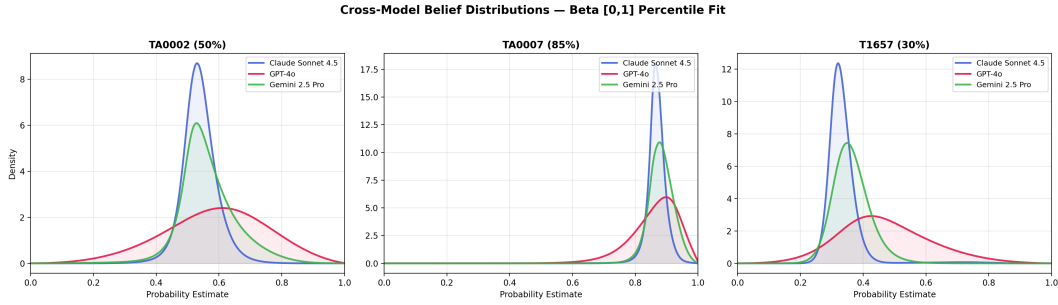


Figure 2: Cross-model distributions show clear separation across all three attack steps.

Results: Median elicited probability tracked baseline approximately linearly (Figure 3), with ceiling effects at high baselines (reduced uplift near 100%). For scenario-level quantity estimates (number of actors, unbounded), the linear relationship persisted without ceiling effects (Figure 4), indicating strong anchoring for both bounded and unbounded estimates.

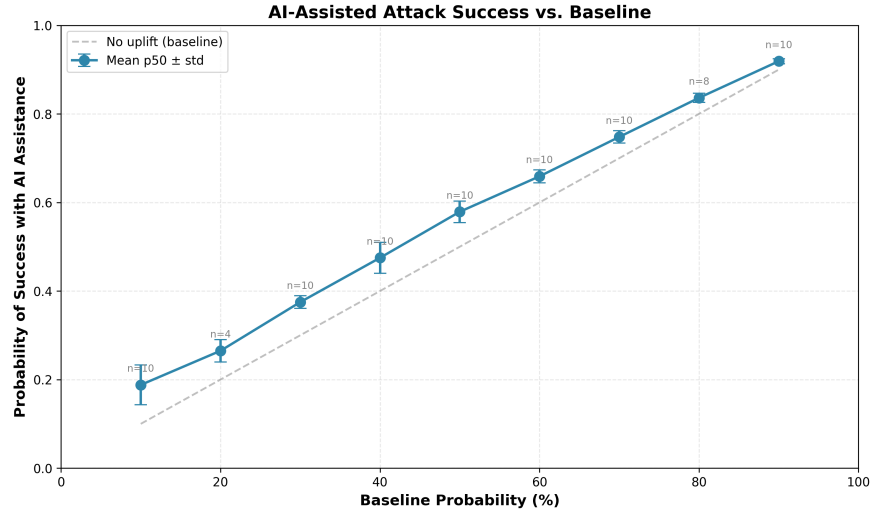


Figure 3: Median probability vs baseline (10–90%), showing approximately linear relationship with ceiling effect. Error bars: IQR across 10 runs.

Prompt Component Sensitivity

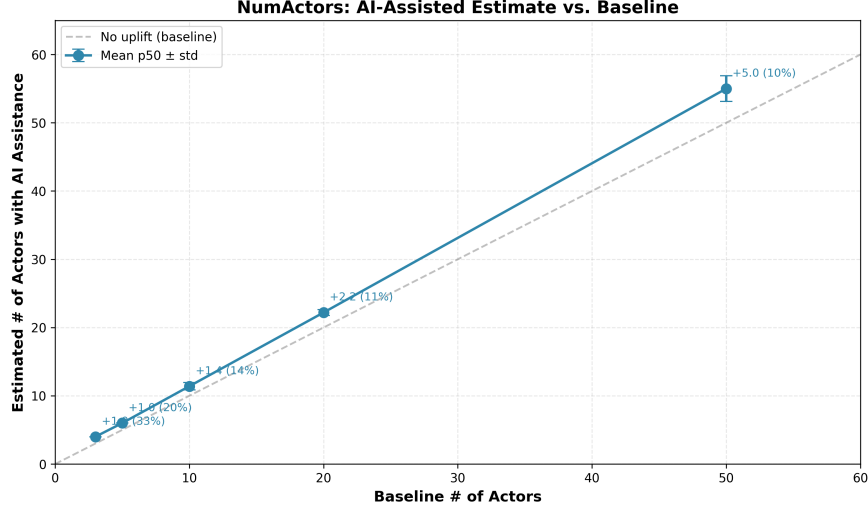


Figure 4: Number of actors vs baseline shows linear relationship without ceiling effects (unbounded estimate).

We tested seven prompt conditions (Claude Sonnet 4.5, single persona, TA0002 50% baseline, Imaginairy, 10 runs per condition) measuring distributional shift via W_1 distance from full control prompt (615 words):

1. **Control:** Full prompt
2. **no_ci:** Removed uncertainty range from baseline
3. **no_baseline:** Removed point estimate from baseline
4. **no_baseline_no_ci:** Removed entire baseline
5. **skip_analysis:** Removed 4-section capability analysis
6. **trim_reasoning:** Removed 3-phase reasoning scaffold
7. **trim_all:** Removed analysis + reasoning + technical output (~ 100 words)

Results (Table 3, Figure 5): Baseline removal caused largest shifts (no_baseline: $W_1=0.154$, $18.1\times$ variance; no_baseline_no_ci: $W_1=0.230$, $12.3\times$ variance). Removing CI alone had minimal impact ($W_1=0.013$, $0.36\times$ variance). Surprisingly, trimming reasoning/analysis reduced variance (trim_reasoning: $0.30\times$; trim_all: $0.07\times$) while maintaining similar central estimates, suggesting convergence under shorter prompts.

Table 3: Prompt sensitivity: W_1 distance from control and variance multiplier.

Condition	W_1	Variance ($\times$ Control)
Control (615 words)	—	1.00 $\times$
no_ci	0.013	0.36 $\times$
no_baseline	0.154	18.1 $\times$
no_baseline_no_ci	0.230	12.3 $\times$
skip_analysis	0.041	0.78 $\times$
trim_reasoning	0.011	0.30 $\times$
trim_all ($\sim$ 100 words)	0.016	0.07 $\times$

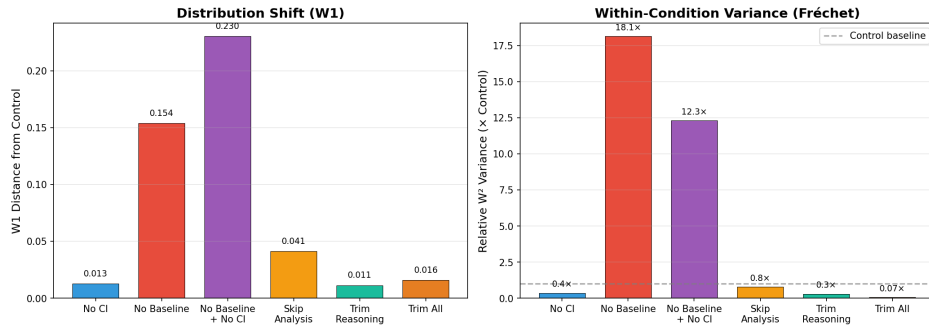


Figure 5: W_1 distance across seven prompt conditions. Baseline-removal shifts most; trimming reduces variance.